# Compass Analytics
## Know Your Customer

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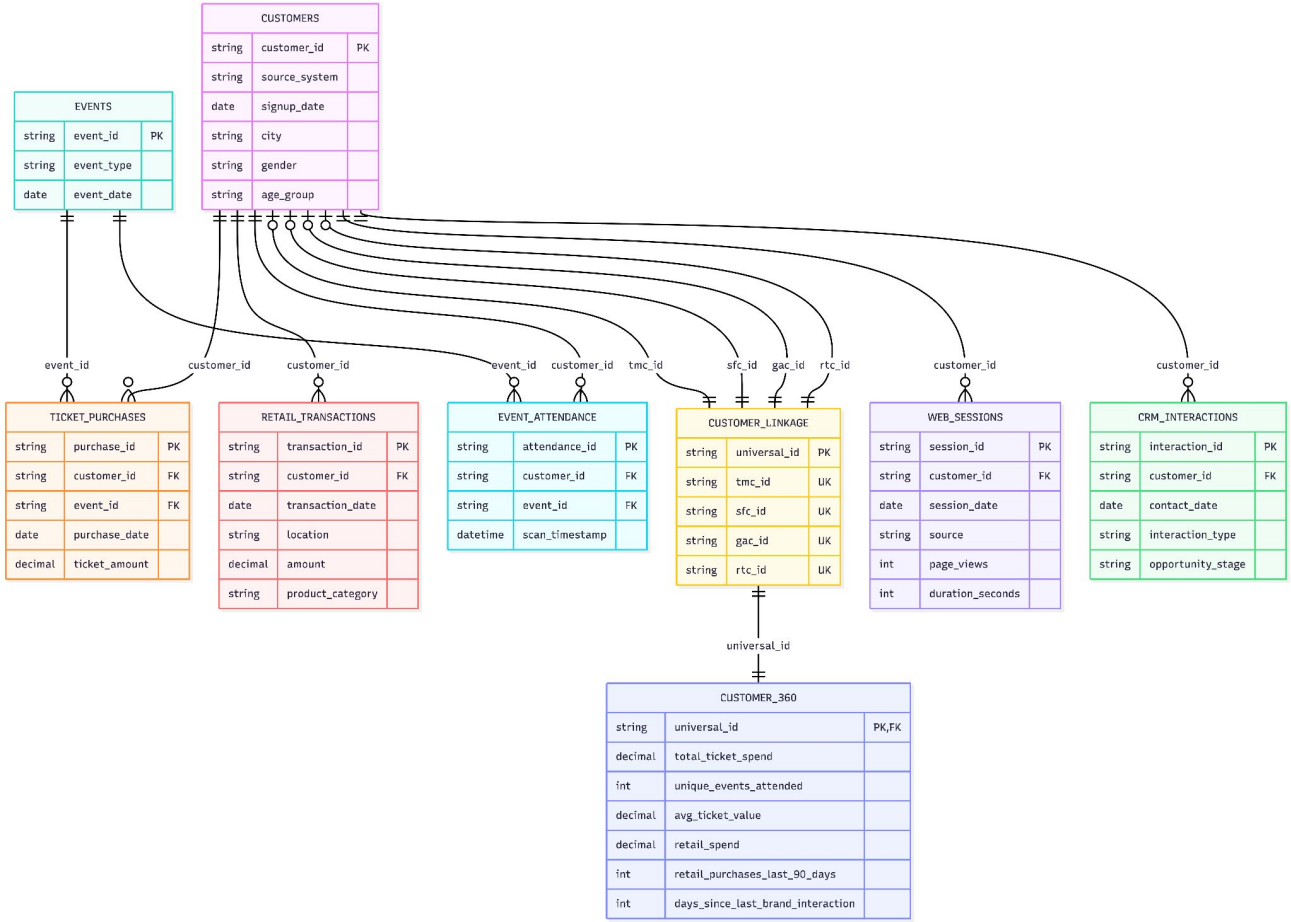
# **Task #1** – Identity Resolution Strategy (ERD)




** Map IDs by source system*

# **Task #1** – Identity Resolution Strategy (Pipeline)

## 1. Source Ingestion

Ingest + Standardize

- Load Ticketmaster, CRM, Web, and Retail data
- Standardize IDs, timestamps, and profile fields
- De-duplicate within each source system
- Identify unique source IDs (~40–55% lack matches)

**Output** → Standardized Source Records

## 2. Matching Sources

Match Across Systems

- Apply deterministic rules on strong identifiers
- Use probabilistic matching for unresolved records
- Score candidate matches by confidence
- Validate existing linkage and mismatches

**Output** → Candidate Customer Links

## 3. Audit Results

Audit + Validate

- Track match logic, confidence, and evidence
- Flag low-confidence or conflicting links
- Monitor quality and cluster consistency
- Create an auditable linkage layer

**Output** → Validate Identity Links

## 4. Resolve Gold Build

Build Unified Output

- Assign synthetic IDs to unresolved records
- Assign a `universal_id` to resolved customers
- Build the `customer_360` gold table
- Generate downstream analytical features

**Output** → Customer 360 Gold Table

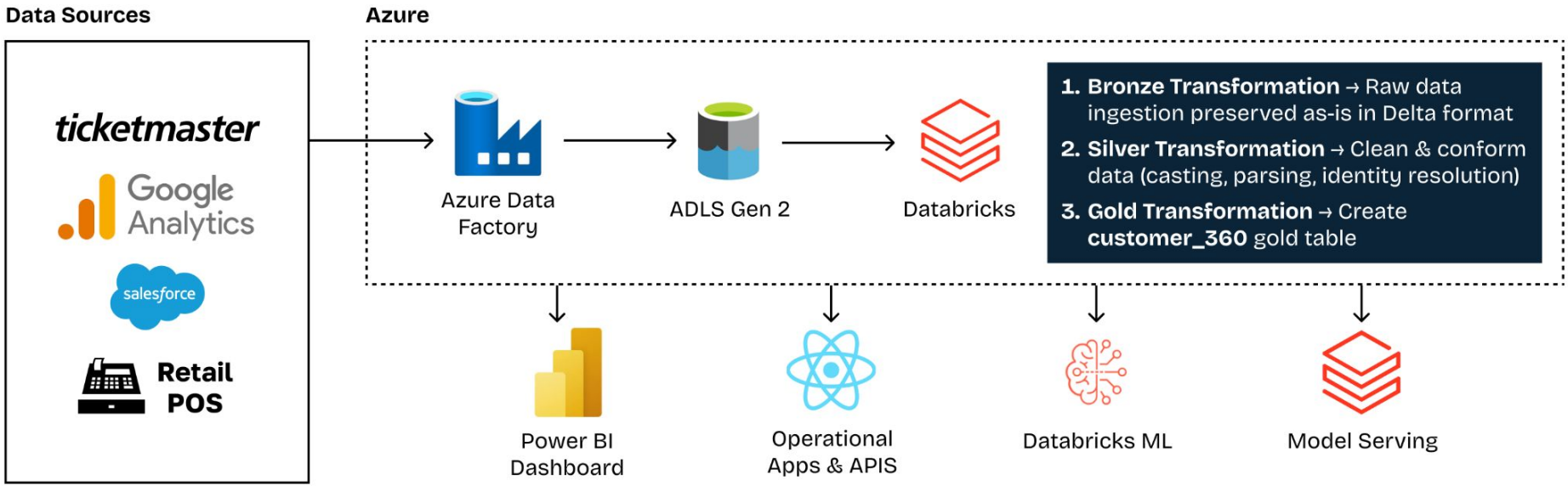
# **Task #2** – Build Customer 360 Gold Table

1. **Unified Customer IDs**
   - Converted source-specific IDs into universal_id using customer_linkage
   - Preserved unmatched records as single-system customers
2. **Standardized Dates and Recency Logic**
   - Parsed all timestamps before feature engineering
   - Used the dataset max date as the recency reference point
3. **Built Required Feature Aggregates**
   - **Ticketing**: Spend, avg ticket value, purchase counts
   - **Attendance / Retail / Web / CRM**: Events attended, retail spend, etc.
4. **Built the Gold Table**
   - Outer-joined all channel features by `universal_id`
   - Added profile fields and days since last brand interaction
5. **Validated and Exported Output**
   - Checked totals, uniqueness, null handling, and feature sanity
   - Exported the final `customer_360` table for downstream use

*Link to Notebook*

*Saved customer_360 to datasets/customer_360.csv*
*511 rows × 26 columns*

# **Task #3** – Data Platform Overview (Architecture)




## Key Points

1. Replace **cron jobs on VMs** with Azure Data Factory scheduled & event-driven triggers

2. Data governance with **Microsoft Purview and Unity Catalog** for role-based access

3. Databricks is **cloud-native, scalable, and can do both ML and AI,** and ADF connects to each source

# **Task #3** – Data Platform Overview (Detailed Overview)

## Platforms

**Azure Data Factory** → Managed ETL service, replaces fragile cron-based pipelines

**ADLS Gen 2** → Cost-effective object storage with ACID transactions via Delta format, enabling refinement

**Databricks Lakehouse** → Unified platform for data engineering and ML with great scalability

**Unity Catalog & Purview** → Enforce access control, lineage, and governance policies

## Data Governance

**Access Control** → Role-based access via Unity Catalog and Microsoft Entra ID so teams only see relevant data

**Microsoft Purview** → Scans ADLS & Databricks assets for data governance issues

**PIPEDA Compliance** → Canadian privacy law requires consent tracking and right-to-deletion

**Data Quality and Monitoring** → Automated alerts on data refreshes, volume anomalies, and pipeline failures

*Microsoft Azure provides managed orchestration, scalable storage & compute, and governance in one ecosystem*